Project Title: Supply Chain Data Analysis Using SQL

1. About dataset

dataset contains 100 records and 24 columns related to supply chain operations.

It includes information about:

Category	Example columns
Products	Product type, SKU, Price
Sales	Number of products sold, Revenue generated
Inventory	Stock levels, Availability, Order quantities
Suppliers	Supplier name, Location, Lead time
Shipping	Shipping carrier, Shipping costs, Shipping times
Manufacturing	Production volumes, Manufacturing costs
Quality	Inspection results, Defect rates
Logistics	Transportation modes, Routes, Costs

Project Objective

The main objective of this project is to analyze supply chain data using SQL to understand product performance, revenue generation, inventory levels, supplier lead times, shipping costs, manufacturing expenses, and product quality. The project aims to extract meaningful business insights from the dataset to support data-driven decision-making, improve operational efficiency, and identify opportunities for cost optimization.

Specific Objectives

1. **Data Exploration:** Explore the supply chain dataset and understand its structure, columns, and records using SQL queries.
2. **Product Analysis:** Identify unique product types, analyze product prices, and find the highest-priced products.
3. **Revenue Analysis:** Calculate total revenue, analyze revenue by product type and SKU, and identify top revenue-generating products.
4. **Sales Analysis:** Calculate total products sold and compare sales volumes across different product types.
5. **Inventory Management:** Identify products with low stock levels to help highlight potential inventory shortages.

6. **Supplier Performance Analysis:** Calculate average lead times for suppliers to compare their delivery performance.
7. **Shipping Cost Analysis:** Evaluate shipping costs across different carriers to identify potential opportunities for logistics cost optimization.
8. **Manufacturing Cost Analysis:** Calculate average manufacturing costs by product type to compare production expenses.
9. **Quality Analysis:** Analyze inspection results and identify products with the highest defect rates to highlight potential quality concerns.
10. **Data-Driven Decision-Making:** Use SQL aggregations, filtering, sorting, and grouping to generate insights that can support better supply chain planning and operational decisions.

Expected Outcome

The project is expected to provide a clearer understanding of supply chain performance by identifying revenue trends, sales patterns, inventory concerns, supplier lead times, shipping expenses, manufacturing costs, and product quality issues. These findings can help businesses identify areas for further investigation and potential operational improvements.